

Ex.No.4

Setting up a Local Area Network (LAN)

5/8/26

Aim:

To design set up and Configure a LAN Connecting four Computers through a switch, Verify Connectivity using ping Command.

Equipment Required:

S.No.	Equipment	Quantity	Specification
1.	Personal Computers (PC)	4	Ethernet (NIC) enabled
2.	Network switch	1	8/16/24 port unmanaged/managed.
3.	Ethernet Cables (Cat5e / Cat6)	4	straight-through, RJ-45 Connectors

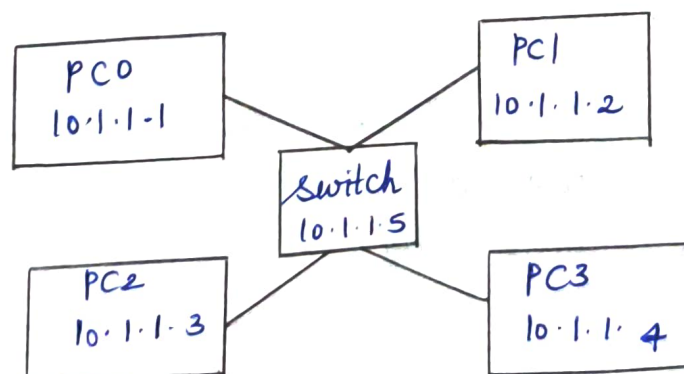
Theory:

A LAN that connect Computers and devices within limited geographical areas.

A star topology, in which end devices Connected to central network switch.

Network Diagram:

LAN was implemented using a star topology.



IP Address Configuration Table:

Device	IP Address	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in Cisco Packet Tracer

Step-1: Place the Devices.

- (i) open Cisco Packet Tracer.
- (ii) From the device category panel, select End Devices and drag 4 PCs onto the workspace.

Step-2: Cable the devices

- (iii) click Connections → select Copper straight-through Cable.
- (iv) click each PC, then click the switch, select FastEthernet.

Step-3: Assign IP Addresses to PCs

- (v) Click PC → Desktop tab → IP Configuration
- (vi) Enter static IP Addresses and subnet Mask for each PC.

Step-4: Configure the switch Management IP:

- (vi) click the switch → CLI tab.

(vi) Commands

enable

Configure terminal

interface Vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

Step-5: Verify Connectivity using Ping

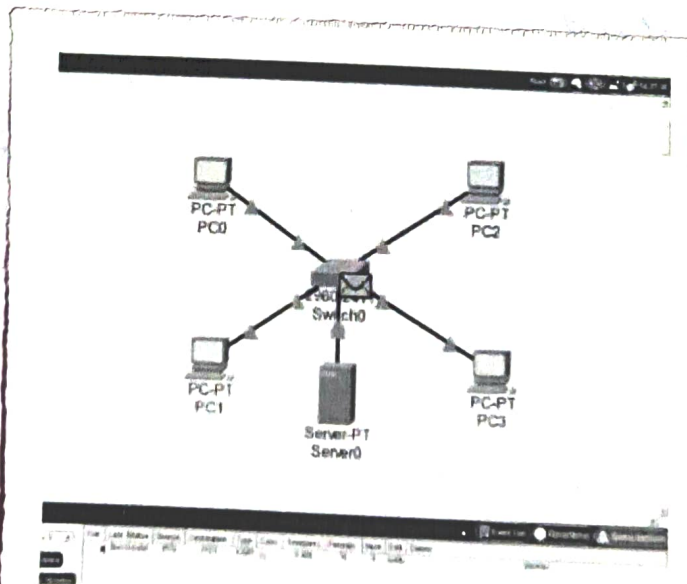
- (vii) Click a PC → Desktop tab → Command prompt.
- (viii) using the ping command to test connectivity to the other PC.
- (ix) successfully "Reply from..." messages Confirm Connectivity.

Step-6: Demonstrate Data sharing

- (x) Add a server device (Enter devices - server-PT)
Connect it to the switch.
- (xi) Assign it an IP address via its Desktop - IP Configuration tab.
- (xii) open the server's services (Config) tab and enable the HTTP service.

Step-7:

- (xiii) switch to simulation mode (bottom right of the workspace).
- (xiv) send a ping between two devices, observing MAC address based forwarding.



Result:

Thus the setting up a Local Area Network (LAN) was implemented and executed successfully.